Magnetic Resonance Spectroscopy

Phil Rodrigues

Jingyi Yen

The Ohio State University

ECE 5206

Professor Clymer

Introduction:

Magnetic Resonance Spectroscopy (MRS) is a technique used in Magnetic Resonance Imaging (MRI) that allows for quantification of the different chemical makeups of a localized area of the imaged tissue. MRS is done in a non-invasive manner with the same setup as an MRI, and is a powerful tool to gather information of the biochemical processes within the body. Primary uses of MRS technology is for identification of clinical conditions, patient monitoring for treatment control, and as a research tool in studying isolated cells, body fluids, and metabolic processes within organs.

Nuclear Magnetic Resonance (NMR):

Like MRI, MRS is a technology based on nuclear magnetic resonance (NMR) principles. This is the physical phenomena that atomic nuclei behave like tiny bar magnets, possessing a quantum property called “spin” which can be thought of as the nucleus of an atom spinning around its own axis. This means, when an external magnetic field is applied, the nucleus behaves as a dipole and aligns along the axis of the applied magnetic field. Nuclei are either polarized parallel or antiparallel with the applied magnetic field (slightly more than half are polarized parallel, since it is the slightly lower energy state), creating the net magnetization vector. The nuclei also spin along their own axes—called precession—with the frequency of this axial spin (ω) being determined by the strength of the magnetic field (B) and the gyromagnetic ratio (γ),

which is unique to each type of nuclei. This relation is expressed by the Larmor equation in (1).

$$\omega = -\gamma B \tag{1}$$

Another external pulse of a radio frequency (RF) electromagnetic signal can then be applied to flip the net magnetization vector to be transverse to the original orientation, also synchronizing the phase of the nuclei precession for a short while. After the RF signal is no longer applied, the net magnetization vector will return the original longitudinal orientation in a spiral manner. This spiral manner induces an electrical signal as described by Lenz's Law, where current is induced by a rotating magnetic field. As the net magnetization vector returns to the longitudinal orientation, more and more nuclei dephase from each other, creating a weaker and weaker induced electrical signal. This process is known as Free Induction Decay (FID), and the delay time is different for different tissues in the body. As seen in Fig. 1, this superimposition of all the resonances of the signals can be decoded with a Fourier Transformation (FT) to obtain a spectrum of the different resonances, and therefore the different substances in a sample [1], [4].

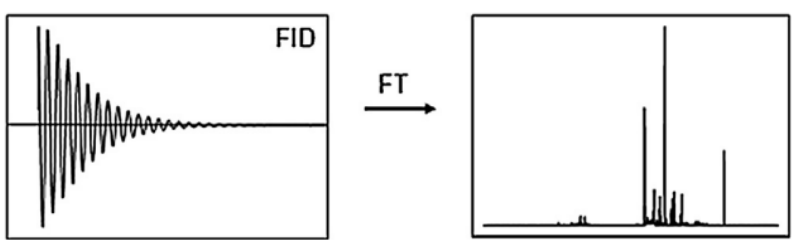

Fig. 1: Fourier transformation from free induction decay to MR spectrum. [4, Fig. 3]

Proton Magnetic Resonance Spectroscopy (1H-MRS):

In order for a nucleus to absorb electromagnetic energy for the spin change, the RF energy has to be at the specific Larmor frequency of that nucleus. Most widely used in MRI and MRS technology is the use of RF signals to excite the resonance of the hydrogen nucleus (1H), also called Proton Magnetic Resonance Spectroscopy (1H-MRS). This is because of many reasons. The Larmor frequency for 1H varies from 60 to 300 MHz from magnetic field strengths from 1.5 to 7.0 T respectively. The human body is relatively transparent to RF at the MHz range, making imaging with 1H resonances accessible. 1H is also the most abundant in the human body, providing many atoms to induce a signal [1].

Additionally, because the specific frequency the 1H nuclei resonants at is a function of the applied magnetic field, the spectrum that is reconstructed is expressed using dimensionless units in reference to a specific material. For 1H-MRS, the reference material is water at 4.7 ppm (parts-per-million) [4].

Chemical Shift:

For a single 1H nucleus, the magnetic field that it experiences is also dependent on local factors. For example, if the nucleus is not free but part of a chemical compound, then the surrounding electrons of the compound will “shield” the external magnetic field. This will shift the magnetic resonant frequency differently depending on the different chemical environments the nucleus is in, which is a function of the chemical structure that the nucleus is part of. This is called “chemical shift” and it is what enables MRS to create a spectrum for different metabolites in the body [1], [4].

The chemical shift (δ), expressed in ppm, given in (2) (from [1. Eq. 2]) is found by the measured resonant frequency (ω) to the reference resonant frequency (ω_{ref}) of a free 1H nucleus as discussed in the previous section. Fig. 2 shows a breakdown of the different spectral signatures of different detectable metabolites in the brain [1].

$$\delta = \frac{\omega - \omega_{ref}}{\omega_{ref}} \tag{2}$$

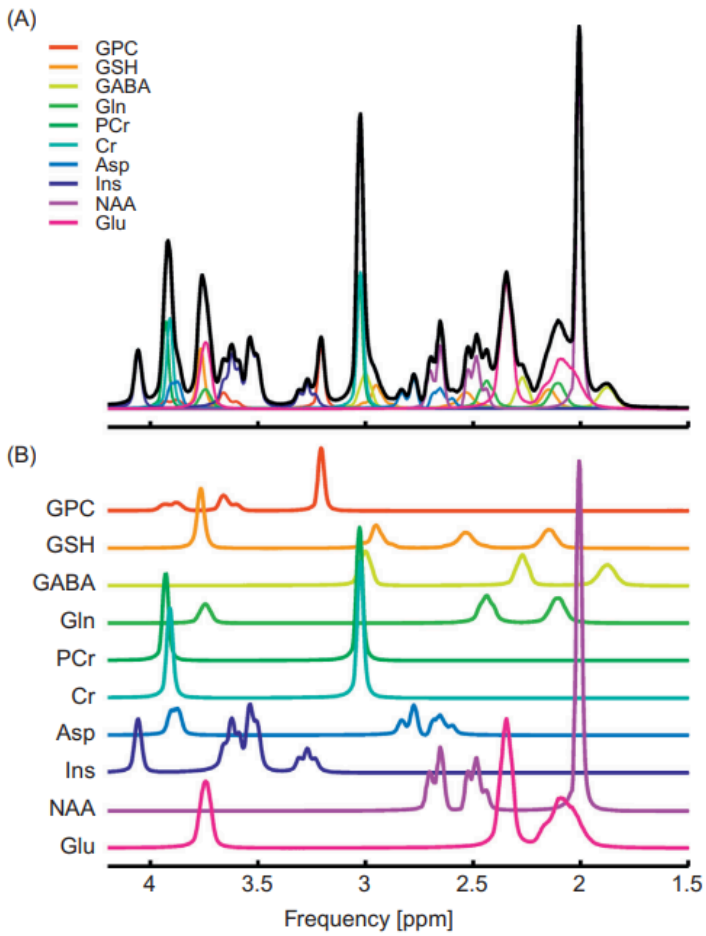

Fig. 2: Quantification of 1H MR spectra as a linear combination of model spectra (LCModel). (A) In vivo 1H spectra consist of a sum of signals from the apparent neurochemical profile. (B) The quantification therefore aims at reproducing the spectrum at hand as a scaled combination of the spectral patterns from individual metabolites. For

each substance, the necessary scaling corresponds to the apparent concentration. [1, Fig. 1.1.2]

Spin-Spin Coupling:

In addition to what chemical environment the nucleus is in, the spins of the neighboring nuclei also can affect the NMR signal. This phenomenon is called spin-spin coupling, or J-coupling. Around half the neighboring nuclei have a parallel magnetic moment to the applied magnetic field, and half have a perpendicular magnetic moment. This creates the NMR signal to have a split peak, with one peak shifted downfield, and the other peak shifted upfield respectively, known as different multiplet structures. This can give even further detailed insight in the molecular structure of the sample.

Spatial Localization, Single-Voxel Spectroscopy:

There is a key distinction between MRI and MRS, specifically when it comes to the localization of the information of the voxels that is being measured from the patient. In MRI, frequency and phase encoded gradients are employed to change the Larmor frequencies of the protons, to be able to determine the position of that voxel. However, if this same technique is applied for localization of signals in MRS, that would interfere with the spectrum that is reconstructed. Now, not only would the differences in the frequencies in the reconstructed spectrum be from chemical shift effects, but also from the frequency encoded gradient. This would not produce an accurate reconstructed spectrum to distinguish different chemical compounds in the sample taken. As a result, different localization techniques not used in MRI are utilized for MRS. Additionally, MRS

is usually applied only when looking at small specific areas where clinicians want more information about, for example for a tumor or a lesion.

As discussed above, when sampling the signal from the precession of the different spins of the 1H nuclei in the patient, there cannot be a frequency encoded gradient so it does not interfere with accurate spectrum reconstruction. However, prior to sampling, slice selection is done similar to MRI. A gradient magnetic field is applied across the entire patient in one axis. This causes different Larmor frequencies of the 1H nuclei along that axis, meaning RF pulses at a certain frequency will only excite nuclei along a specific slice. This slice selection is then repeated two times in orthogonal directions. These three RF pulses at specific frequencies along the gradient magnetic fields among three different axes create first a slice, then a column, then a single voxel typically from 4-8 cm^3 for localization [5].

The three RF pulses for MRS follow the similar MRI imaging spin-echo technique. The first RF pulse is a 90 degree pulse in one axis, followed by a 180 degree rephasing pulse in an orthogonal axis, finished by another 180 degree rephasing pulse in the last orthogonal axis. The echo of these spins is then sampled at time TE. This sequence for single voxel localization of signal is known as Point-RESolved Spectroscopy (PRESS) [1], [5]. A visualization of the PRESS sequence is given in Fig. 3.

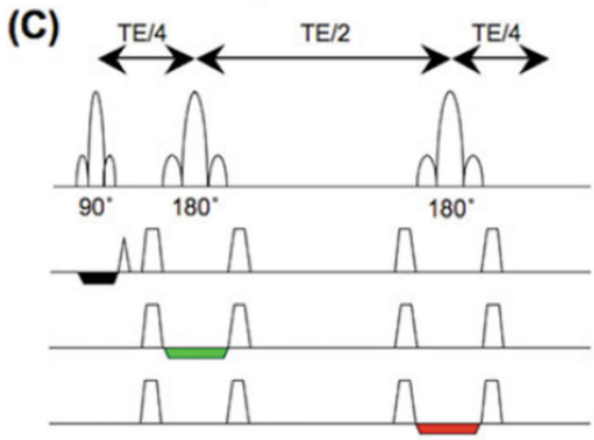

Fig. 3: The PRESS sequence, consisting of a slice-selective 90° excitation pulse and two 180° slice selective refocusing pulses. [5, Fig. 9.2 (c)]

Another sequence commonly used is STimulated Echo Acquisition Mode (STEAM). The concept is the same as PRESS, but instead of a 90 degree, 180 degree, and 180 degree RF pulse sequence, three 90 degree RF pulses are used, simulating an echo [1], [5]. A visualization of the STEAM sequence is given in Fig. 4, with a visualization of both of these gradient based single-voxel localizations shown in Fig. 5.

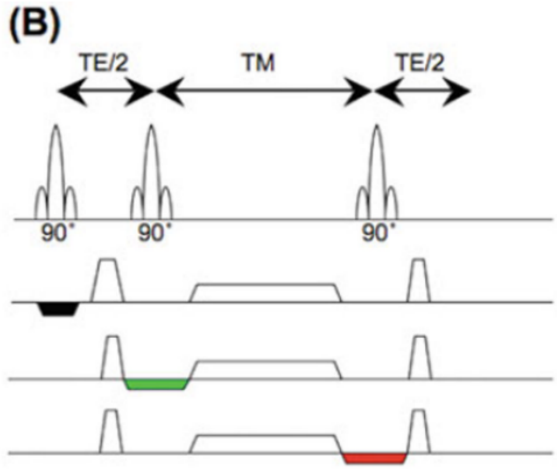

Fig. 4: The STEAM sequence, consisting of three 90° slice-selective pulses. [5, Fig. 9.2 (b)]

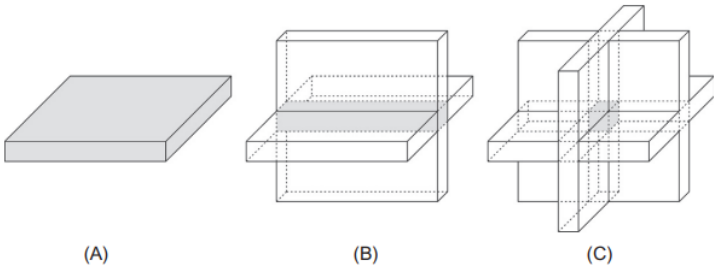

Fig. 5: Concept of gradient-based localization for single-voxel MRS methods such as STEAM and PRESS. The combination of three slice-selective RF pulses reduces the sensitive volume to a slice (A), a column (B), and ultimately a voxel (C) as intersection of the three slices. The method allows region-specific MRS, since the acquired spectral signals stem from the selected volume only. Magnetization that is located outside the selected voxel and therefore has not been excited by all three RF pulses is eliminated by additional "spoiler" gradients (not shown). [1, Fig. 1.1.4]

For different metabolites and compounds in the body, there are different relaxation or T2 times. This is a measure of how long it takes for a nucleus to decay from precessing (and therefore emit a RF signal to be measured) after an external RF pulse is applied. This means that depending on the TE time of the sequence, some signals from compounds might have already decayed and cannot be measured. As a result, sequences with lower TE's are desirable. However there is a tradeoff, the shorter the TE, the more baseline distortion there is, allowing for more difficult analyzation [2]. This can be seen in Fig. 6.

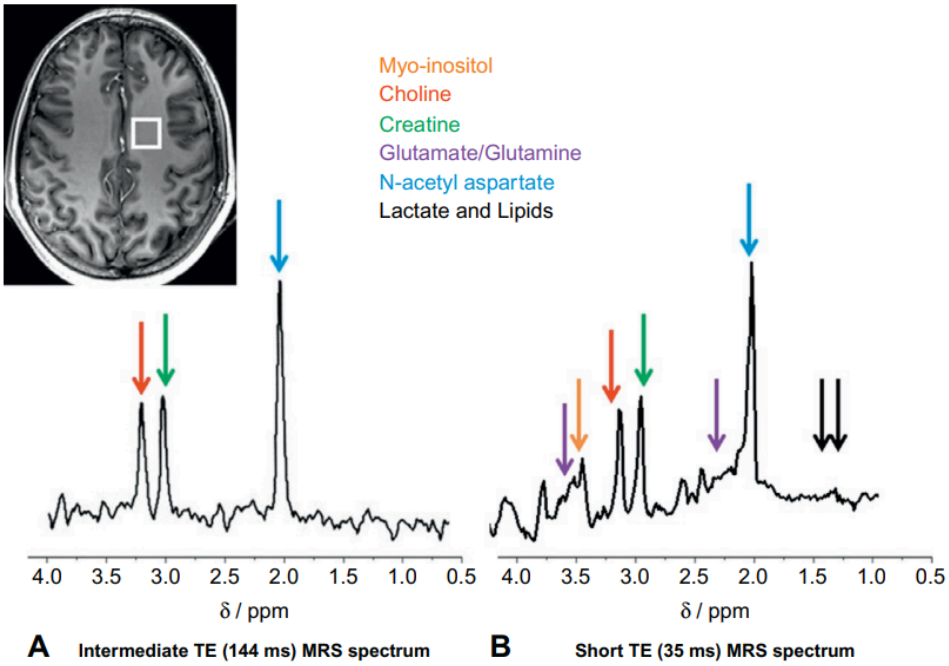

Fig. 6: Magnetic resonance spectroscopy (MRS) of white matter in a normal brain. (A) Intermediate echo time (TE) spectra of 135–144 ms have less baseline distortion and are easy to process and analyze but show fewer metabolites than short TE spectra. (B) Short TE demonstrates peaks attributable to more metabolites, including lipids and macromolecules, glutamine and glutamate, and myo-inositol. [2, Fig. 5.1]

Some comparisons between PRESS and STEAM is that TE time for STEAM is shorter than that of PRESS, all while having a lower RF power requirement. However, because PRESS uses 180 degree rephasing and refocusing pulses, the signal can theoretically be twice the strength as that from STEAM [2], [4], [5].

Spatial Localization, Multiple-Voxel (MRSI) Spectroscopy:

A limitation of Single-Voxel Spectroscopy is just that; only one voxel can be looked at a time. While frequency encoding is unsuitable for spatial resolution due to its interference with the reconstructed spectrum, phase-encoding can be applied in two

directions before data acquisition, typically resulting in the creation of a 2D slice comprising voxels with spectrum information. This can measure multiple voxel locations at the same time, at the cost of longer overall scan times. This technique is called Magnetic Resonance Spectroscopic Imaging (MRSI), or Chemical Shift Imaging (CSI) [1]-[5]. A MRSI slice can be seen in Fig. 7.

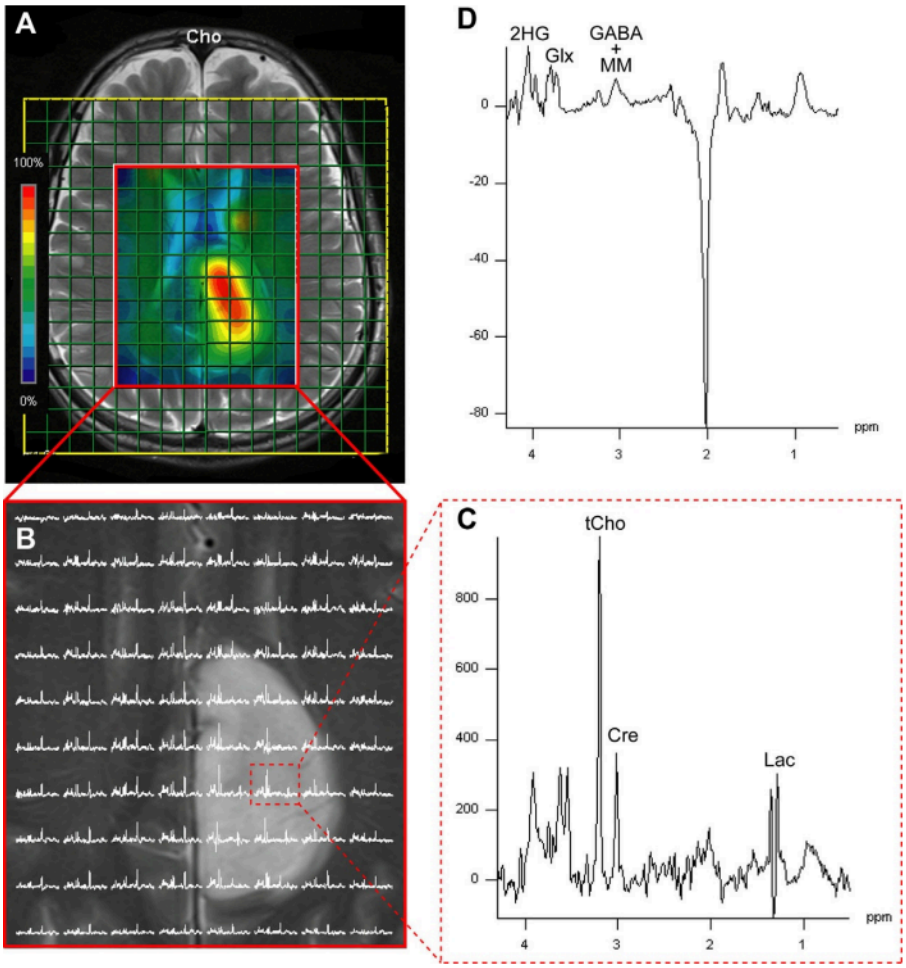

Fig. 7: Magnetic resonance spectroscopic imaging (MRSI) of a patient with preoperative glioblastoma multiforme (GBM) shows elevated choline (Cho: A–C). Spectral editing MRS (D) reveals 2-hydroxyglutarate (2HG in panel D). This patient proved to have an R132G isocitrate dehydrogenase (IDH) mutation. GABA, gamma-aminobutyric acid;

Glx, glutamate/glutamine; MM, macromolecules; tCho, total choline; Cre, creatine; Lac, lactate. [2, Fig. 5.7]

There are other techniques to speed up the acquisition of MRSI data, lumped under the umbrella term of “Fast MRSI” techniques, but that is beyond the scope of this paper [1], [2], [4], [5].

For both Single-Voxel and Multiple Voxel Spectroscopy, smaller sized voxels mean longer data acquisition times because of multiple samples needing to be averaged to average out the noise to zero, and improve the signal-to-noise ratio (SNR) [1], [2], [4], [5].

Water Suppression:

1H was the chosen molecule in which to select the RF pulses to excite at the Larmor frequency in order to receive a signal back due to NMR properties. A problem with this approach is that water will have a high resonance with this approach. The human body, being made of more than 70% water, is in a much higher concentration than other detectable metabolites (10^3 - 10^4 times smaller concentration), making other metabolites very hard to detect [1].

Different approaches have been developed to suppress the signal from water, with the most common being the CHEmical Shift-Selective (CHESS) method. This method involves using a frequency-selective RF pulse to excite the water resonance, then using “crusher” or phase-spoling gradients to minimize this excited water resonance. This does not minimize any of the signals from the metabolites that are actually trying to be measured. CHESS is often applied prior to the actual MRS, since

there is no interaction with the metabolites. CHESS can be repeated multiple times to further calibrate and remove unwanted water resonance [1], [5]. This can also be done during a PRESS sequence as shown in Fig. 8 with a 2D-PRESS-MRSI pulse sequence. The effects of water suppression on the resulting reconstructed spectrum is shown in Fig. 9.

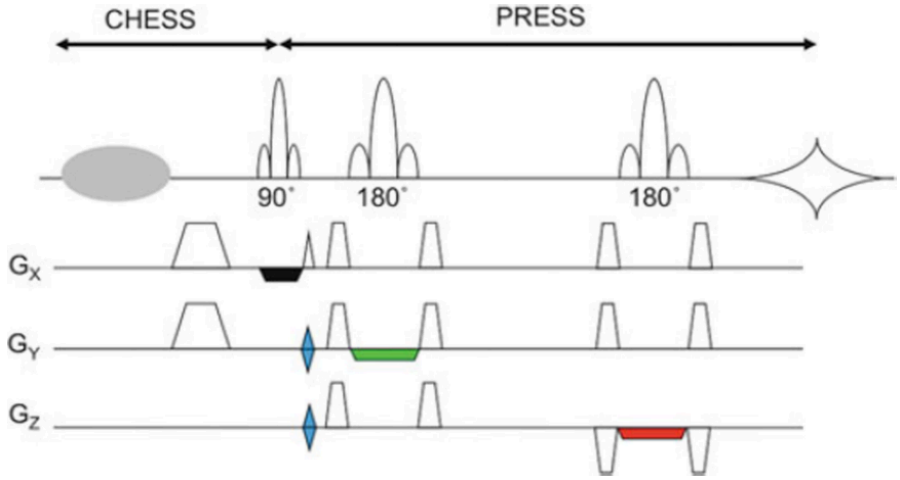

Fig. 8: 2D-PRESS-MRSI pulse sequence. A PRESS sequence is used to excite a large volume of brain tissue while excluding signal from lipid in the scalp and/or regions of poor field homogeneity, and then phase-encoding gradients (in blue, GY and GZ) are used to localize spectra from regions within the excited region. A CHESS prepulse and crusher gradient are applied for water suppression. Crusher gradients applied around the 180° refocusing pulses are also shown. [5, Fig. 9.3]

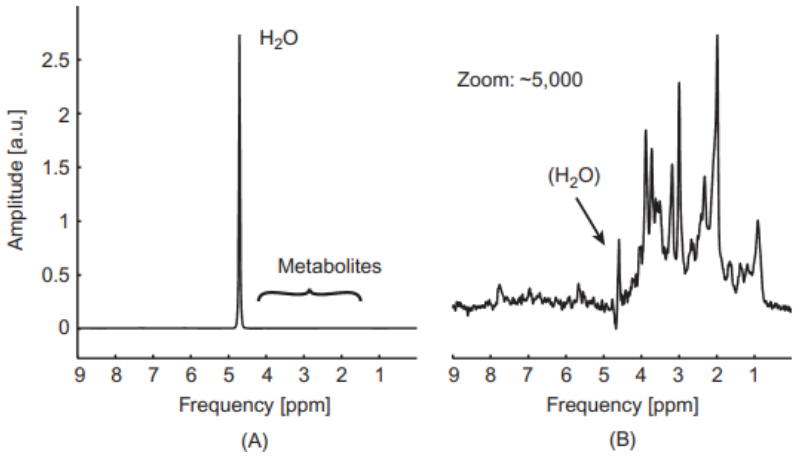

Fig. 9: The role of water suppression for in vivo 1H MRS of the human brain. (A) The water content in human brain tissue largely exceeds the concentrations of other observable substances and therefore dominates the appearance of the spectrum. (B) Suppression of the water signal allows an unobstructed view of the neurochemicals of interest. [1, Fig. 1.1.1]

Lipid Suppression:

One of the most common uses of MRS is for measuring the different concentrations of different metabolites and compounds while imaging the human brain. As such, there are other substances in the head that contribute to unwanted signals that would crowd out the smaller in concentration, less detectable metabolites. Lipids in scalp tissue are one such undesirable substance in high concentration. The resonances that come from these lipids in the scalp can have signal strengths that are 100-1000 times stronger than other brain metabolites [1], [5].

The most common technique to suppress the signal from the lipids in the scalp is called Outer Volume Suppression (OVS). This technique is similar to CHESS, in that the signal to be suppressed is first intentionally excited, then minimized or erased using

phase-spoiling. The distinction for OVS is that frequency selective RF pulses are used to excite entire planar slices, similar to the slice selection used for MRI. These excited slices are then minimized using phase-spoiling [1], [5]. Visualization of the slice selection technique of OVS is given in Fig. 10, with an example sequence given in Fig. 11.

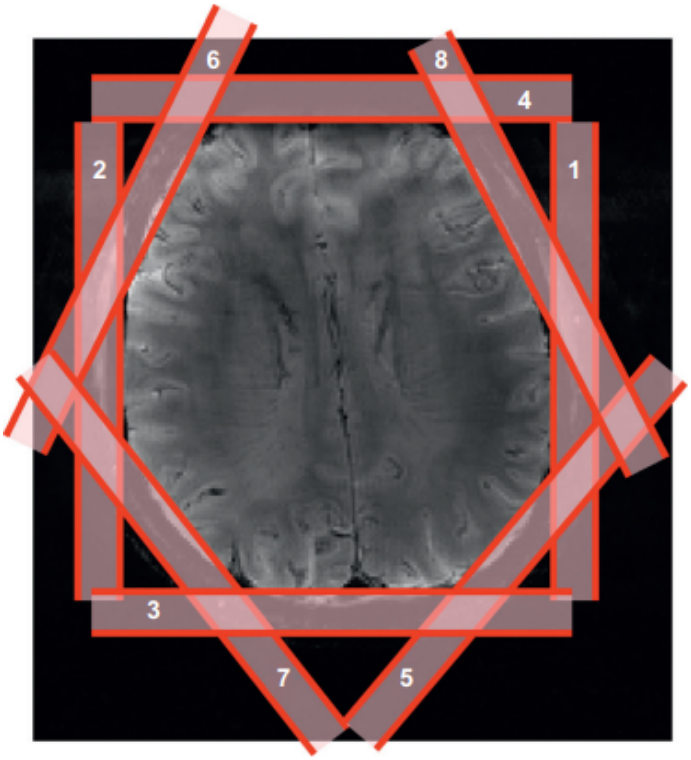

Fig. 10: OVS for MRSI of the human brain. Signals from cranial lipids can be minimized by their selective excitation followed by phase spoiling gradients to eliminate them. The oval shape of the skull in an axial section through the human head is hereby typically approximated by a series of planar OVS slices. [1, Fig. 1.1.6]

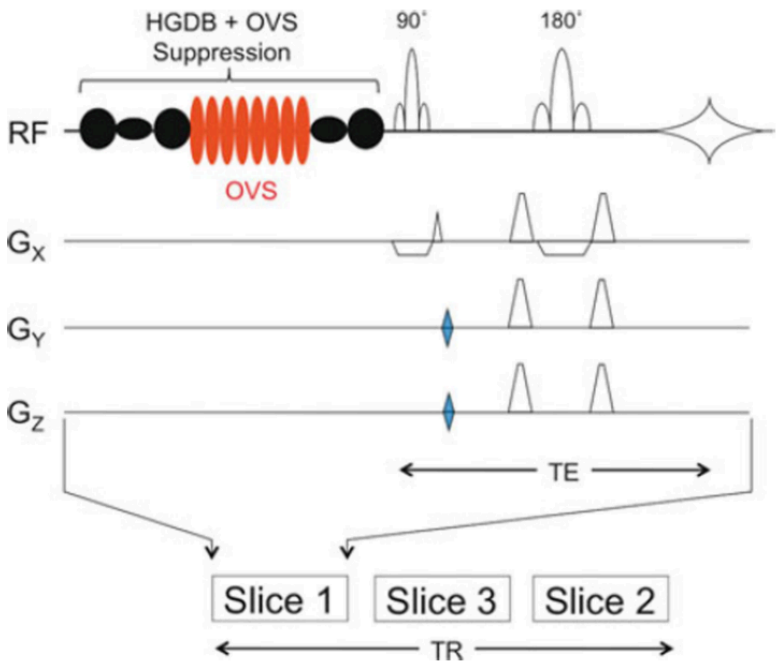

Fig. 11: A multi-slice 2D-MRSI pulse sequence. A slice-selective spin echo is preceded by an optimized water and lipid suppression scheme ("HGDB"), including outer-volume suppression (OVS) pulses (8, indicated in red) for spatial suppression of scalp signals. 2D phase-encoding gradients (blue) are applied on GY and GZ. In this example, three slices are collected within one repetition time (TR). Gradients associated with the water and lipid suppression are omitted for clarity. [5, Fig. 9.5]

Techniques to Combat Magnetic Field In-homogeneity:

For an undistorted reconstructed MR spectrum, it is essential for the magnetic field across the measured sample to be homogeneous. This ensures that any frequency differences observed in the sample result from chemical shifts due to distinct compounds rather than variations in the applied magnetic field. If a magnetic field is not homogeneous across a sample, then for a single chemical compound, the spectral peak

will not be at a single frequency, but spread and broadened [1], [2], [4]. This is a loss of sensitivity, spectral resolution and precision. Even though the electromagnet is extremely strong, it alone cannot guarantee a perfectly uniform magnetic field across a sample.

As a result, a technique called “shimming” is applied. Small shim coils (small wire coils) are placed around the sample, and a current is run through these shim coils to induce a small local magnetic field [1], [2], [4]. This is to supplement the main magnetic coils in creating a homogenous magnetic field.

Tuning and locking is another technique applied to creating a homogeneous magnetic field. This technique is simply continually monitoring a specified signal (deuterated hydrogen, or a stable isotope of hydrogen), and constantly adjusting the field by tiny amounts to preserve the same Larmor frequency [4].

Alternative Nuclei and Isotopes to be used in MRS:

The 1H nucleus was chosen for the one for which the RF excitations would match the Larmor frequency in order to get the NMR effects for the imaging. However, there are other MR-detectable nuclei of biochemical relevance and sensitivity in the body and brain. 31P MRS serves as an alternative method to image brain metabolism and investigate related disorders, while 13C spectroscopy is utilized for studying conditions like Alzheimer's, Canavan's disease, or epilepsy. Additional types of nuclei used in MRS include 23Na, 19F, and 17O [1]. For some diseases, it is helpful to utilize a multi-nuclear MRS approach as a powerful tool to further assess multiple aspects of a disease.

Brain Tumor Characterization:

One of the major ways in which MRS has been applied in the medical field is in the characterization of brain tumors. Specifically, in diagnosis, neoplastic tissue typically exhibits elevated Choline (Cho) levels and decreased N-Acetyl Aspartate (NAA) levels, as can be seen in Figure 12. Elevated lactate and lipids are also typically seen with high-grade neoplasm tissues, and even low-grade tumors can be diagnosed with moderate elevations in Cho levels. Doctors are also able to study the gradient decline of Cho levels in order to distinguish between primary central nervous system neoplasm and metastasis, with more abrupt decline representing metastasis as opposed to a gradual decline [2].

Fig. 12: Higher levels of choline (Cho)/N-acetyl aspartate (NAA) can be seen in high-grade glioma compared to anaplastic astrocytoma. GBM, glioblastoma multiforme. [2, Fig.5.5]

MRS can also be used in the processes of therapeutic planning and prognosis, as doctors can utilize 2D or 3D magnetic resonance spectroscopic imaging (MRSI) in order to identify the most malignant regions of a tumor, leading to more effective treatment planning. Similarly, prognoses on survival can be made by studying mutations with the enzyme isocitrate dehydrogenase (IDH), as when individuals have mutations in the IDH, accumulation of metabolite 2-hydroxyglutarate can be detected in spectral editing. Tumor patients who are identified to have mutations in the IDH-1 tend to survive approximately twice as long as those who do not, illustrating how MRS can lead to significant tumor prognosis. MRS is also heavily utilized in monitoring treatment response, as it can determine radiation necrosis through slight elevation of Cho levels and presence of lipid resonance, and can also detect recurrent tumors through high Cho levels. MRS also presents the ability to measure NAA/Cho levels as an imaging biomarker to assess the response of antiangiogenic treatment, with indications such as increase of NAA/Cho levels 8 weeks post treatment indicating 6 month progression free survival [2]. Being able to quickly and efficiently identify tumors and determine their malignancy is a crucial ability that positively impacts the survival outcomes of millions of victims.

Metabolic Disorders:

MRS also plays a major role in the indication and diagnosis of metabolic disorders due to the many biomarkers in MRS indicating each. Either based on the accumulation or absence of metabolites, doctors are able to indicate leukodystrophies, mitochondrial disorders, and enzyme defects. Enzymatic diseases such as maple syrup

urine disease can be detected by a high accumulation of branch chain amino acids and oxo acids, illustrating the body's inability to break these acids down properly. Similarly, disorders such as phenylketonuria and nonketotic hyperglycinemia can be recognized through high concentrations of phenylalanine and glycine (Gly) respectively, as well as leukodystrophies such as Canavan's disease which are recognized from abnormally high NAA levels [2]. The importance of being able to identify metabolic disorders can be illustrated through figure 13, which depicts the MRS results of a 2-year-old boy. As can be seen in his first MRS data, his cerebral creatine (Cr) levels are extremely deficient, displaying enzymatic defects in his creatine transport from the liver and pancreas. After supplementing with creatine monohydrate for 3 months, the new MRS data on the boy shows a much higher and normal reading for cerebral creatine levels, emphasizing the impact of MRS on diagnosis and treatment. A doctor without this technology would not be capable of looking at an infant presenting with specific symptoms and diagnose him with "creatine deficiency", and by the time they would be able to figure that out, it would possibly be too late. MRS allows for diagnostic excellence and quick treatment solutions.

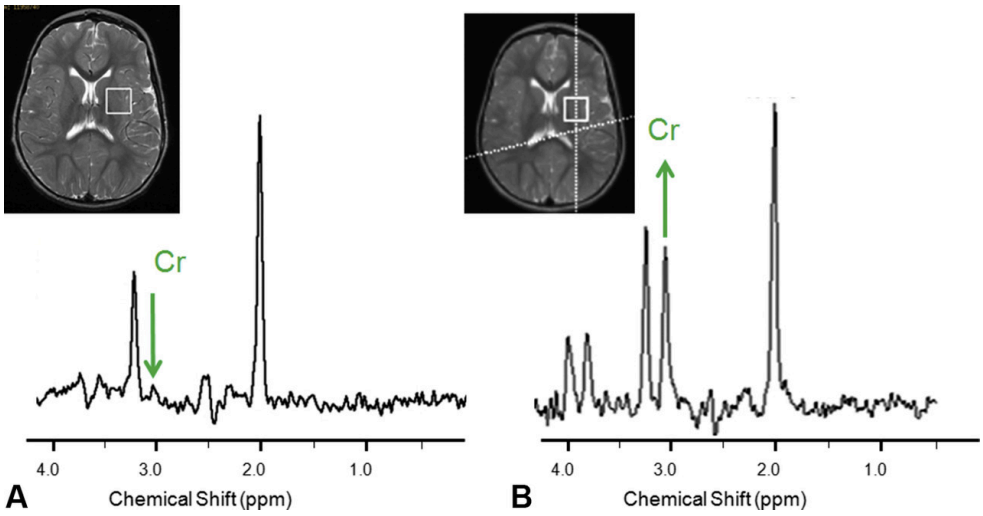

Fig. 13: Magnetic resonance spectrum of a 2.5-year-old child with creatine (Cr) deficiency before and after 3 months of supplementation of creatine monohydrate. [2 Fig.5.11]

Brain Infections and Traumatic Brain Injuries:

Given the nature of brain infections and traumatic brain injuries, these events need to be diagnosed and treated upon as soon as possible for chances of survival. MRS can be useful to diagnose and differentiate between different types of infectious lesions in the brain in a much shorter time than that of laboratory sample testing. Pyogenic abscesses are able to be recognized in MRS through the presence of cytosolic amino acids in collected data such as valine, alanine, leucine, lactate, acetate, and succinate. Common metabolites to show up on the spectrum for typical brain tissue, such as NAA, Cr, and Cho will not be present on the results of that of pyogenic abscesses. Other types of cysts and abscesses, such as tuberculosis abscesses, will only show the presence of lactate and lipid signals, with amino acids such as the cytosolic amino acids listed previously not being present. Some may only show the presence of a few amino acids, such as parasitic cysts only showing succinate and acetate [2]. Due to the wide range spectral results of different types of abscesses, MRS is a quick, non-invasive and effective way to distinguish between each and treat the infections accordingly.

In traumatic brain injuries, while MRS cannot directly assist in the immediate treatment of such injuries, it can play a major role in determining the future outcome of the victim. Utilizing biomarkers such as the NAA/Cr ratio can help predict the recovery

outcomes of a given patient, as with decreased NAA/Cr ratio, the poorer the future neurological outcome usually is for the individual. Likewise, individuals who displayed an elevated mean Cho post-injury or lactate presence also tend to have poorer outcomes than individuals that don't, as these biomarkers illustrate potential inflammation presence in the brain. This shows the value that MRS can present during the recovery period and whether or not the victim is trending towards cognitive deficit/disability or remission. Hypoxic ischemic injury can commonly happen after traumatic brain injuries, and even generally through means of heart failure, stroke, drowning, etc.. Decreased NAA and increased lactate and Cho are also major indicators for unfavorable outcomes in hypoxic ischemic injuries as well, with higher glutamate and glutamine (Glx) levels indicating higher injury severity [2]. In this way, MRS can also indicate how to approach treatment for events such as heart attack and stroke and determine the efficacy of these treatment processes, positively impacting the recovery processes for millions yearly.

Neurologic and Neurodegenerative Diseases:

MRS also plays a significant role in the field of neuroscience, specifically in the fast and effective diagnosis and prediction of neurodegenerative and neurological diseases. Alzheimer's disease is the most prevalent neurodegenerative condition in society. Individuals undergoing MRS testing before showing symptoms consistently exhibit decreased NAA/Cr ratios alongside increased mI/Cr and Cho/Cr ratios. These changes serve as indicators of the impending onset of Alzheimer's. Moreover, a direct relationship has been established between the extent of NAA reduction and disease

severity, showcasing the remarkable diagnostic and predictive capabilities of MRS. Other significant neurodegenerative diseases such as Parkinson's and MS had originally not shown MRS to be significantly helpful, but with research, even these disorders are now being effectively addressed with MRS. Performing MRS specifically in the lentiform nucleus and striatum parts of the brain have shown an identifiable decrease in NAA in Parkinson's victims, a biomarker that was previously insignificant in the detection of Parkinson's. Similarly, while the acute phase of MS plaques would illustrate decrease in NAA and Cho elevation, biomarkers would return to normal after the acute phases, making long-term prediction of MS not available. But, over time, scientists noticed a slight permanent decrease of NAA and increase of Cho in MS victims after experiencing significant acute episodes, setting up MRS as a great indicator of neuronal damage and future outlooks of long term MS [2].

MRS is very effective in diagnosing neurological disorders such as epilepsy and epileptic seizures. While epileptic seizures can commonly be attributed to mitochondrial disorders which can be diagnosed by studying the metabolites discussed in the Metabolic Disorders section, sometimes epileptic seizures can also be attributed to mass growth in the brain as seen in Figure 14. While MRS's assistance in diagnosing epilepsy due to mitochondrial disorders is obvious, its assistance in diagnosing growths which can be found on MRI may be less intuitive. While both cases in Figure 14 may look the same, the top case represents a neoplastic growth and the bottom case represents a dysplastic growth. As can be seen, the neoplastic growth can be identified by increased Cho, decreased NAA, and lactate presence, while dysplastic growth only shows very minor NAA decrease with constant Cho and no lactate presence. MRS

being able to distinguish between the two, is critical both for the treatment process and positive future outcome of the patients, as neoplastics illustrate uncontrolled growth while dysplastics are comparatively tame. The two individuals may require drastically different treatment processes in order to effectively reach remission, and without MRS technology, one may approach both of their treatment processes the same way, lowering both of their chances of survival.

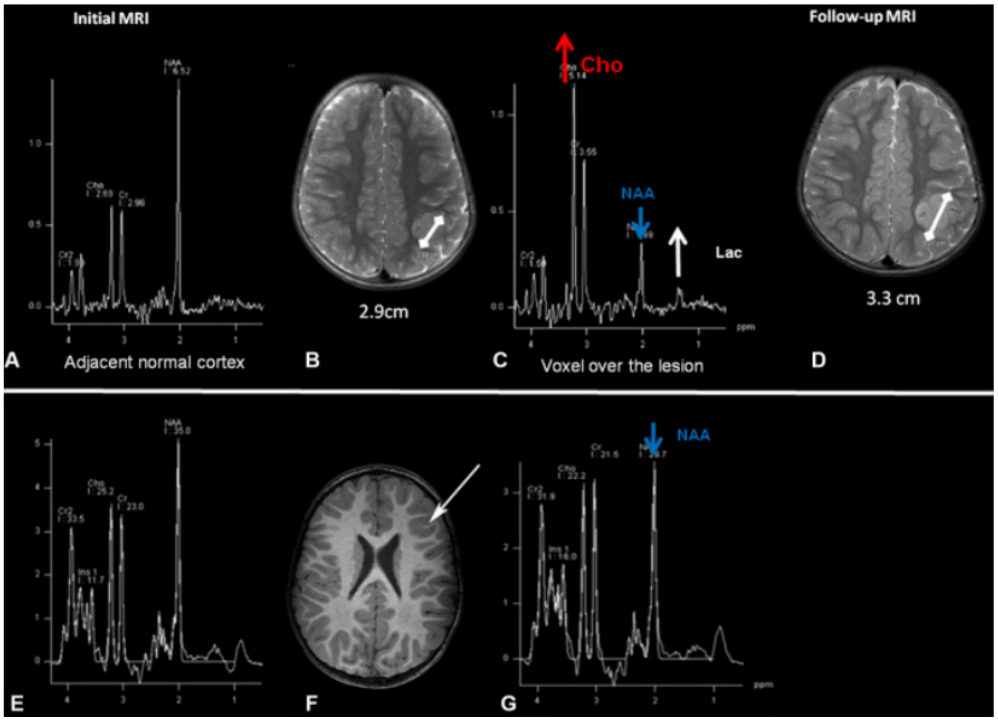

Fig. 14: Magnetic resonance spectroscopy (MRS) of a neoplasm (top) vs a focal cortical dysplasia (bottom).[2, Fig. 5.12]

Psychiatric Disorders:

Attributed to the modern capabilities of MRS, neurotransmitters such as glutamate (Glu), glutamine (Gln), and γ-Aminobutyric acid (GABA) are able to be measured and accredited to specific psychiatric disorders. This allows for doctors to be

able to detect altered neurotransmission within an individual's brain, leading to a true physical analysis of why the individual's brain operates differently. Several mood disorders and major disorders such as Schizophrenia have been attributed to decreased GABA and Glu levels tied with changes in Cho and Cr levels, leading to what were once subjective diagnoses' to be physically quantified. Other "subjective" diagnoses that are much more apparent in society such as OCD, bipolar disorder, and anxiety disorders have been attributed to level changes in NAA, Cho, and Cr [2]. This illustrates a major scientific discovery and promises a strong future in psychiatric diagnosis as by knowing what deficiencies/abundances of neurotransmitters and metabolites lead to specific disorders. Treatments could then be developed to legitimately alter brain chemistry of patients for long term improvement to their disorders.

One major trend in the psychiatric disorder area of research with MRS is the study of the impact of neurotransmitters and metabolites on substance use disorders (SUDs). By being able to attribute the role of these biomarkers on those who are prone to substance abuse, potential treatments can be discovered to assist those who suffer from these addictions to no longer desire the substances which are destroying their bodies. Cognitive deficits which result from substance use disorders have been attributed to impaired creatine kinase activity (CK), which consequently limits the amount of available adenosine triphosphate (ATP) in the brain, which is necessary for high level cognitive operation. Neurotransmitters such as Glu and GABA are also emerging as having a significant impact on those who are easily addicted to substances, and with more application of advanced MRS methods in the near future,

these attributions will be able to be more selectively quantified, but currently these findings are lacking in human SUD research [6]. MRS is promising in finding treatments to many SUDs and even their cognitive consequences after long term use.

Cognitive Function and Brain Connectivity:

The major area of research and advanced study which applies MRS lies in the research of cognitive function and brain connectivity. Due to the extremely high effectiveness MRS provides in detecting neurotransmitters and metabolites in brain tissue, MRS is one of, if not the very most advanced technology that researchers are able to use to study the brain's physical operation, the most complex and least understood organ in modern science. MRS is able to contribute to the decision-making process of the brain and not only determine why a certain individual makes the decisions they do, but can also determine successful measures to alter the decision-making process of the individual chemically. Not only this, but MRS provides great insights into avenues of human attention span, also evident in its efficacy in diagnosing attention disorders such as ADHD. Memory, another significant process of the human brain, is also able to be more understood through MRS research (MRS plays a significant role in diagnosing diseases such as Alzheimer's as mentioned previously). Thanks to the wonders able to be achieved by MRS, neurotransmitter analysis only becomes easier and researchers are constantly learning more about how neurotransmitters and metabolites interact with each other in the brain and how these interactions impact cognitive function [2, 6]. In doing so, human understanding of brain

connectivity, decision-making, attention, and memory operations are being advanced at an exponential rate.

Pharmaceuticals:

MRS is able to utilize NMR processes to effectively contribute to the pharmaceutical industry as well. By performing NMR processes on drugs, pharma researchers can effectively break down all of the present substances within the drug, being able to detect impurities and over/under abundances of certain compounds. This allows for researchers to quality control drugs simply by analyzing NMR spectral data of drug contents, making it not only fast and easy but also accurate and safe for consumers. Imagine if by accident, a batch of “off the shelf drugs” accidentally had the presence of an impurity in lethal amounts to the user? Technology such as MRS is able to quickly and successfully identify accidents like these before they even reach the shelves and without damaging products due to its non-invasive nature, making MRS not only critical in diagnostic settings but also in quality control environments.

MRS/NMR also is critical in the sector of drug discovery, by studying biomarkers in the human brain and their responses to trial treatments. MRS has majorly contributed to drug discoveries for neurodegenerative diseases such as Alzheimer’s disease, dementia, and MS, diseases which have evaded a sure-fire solution for decades. MRS is also extremely useful in oncology in examining the efficacy of cancer treatments such as Avastin (bevacizumab) and limb salvage treatments, monitoring their impact on critical biomarkers. As can be seen in Figure 15, MRS was utilized to analyze a patient with a cancerous brain tumor both before and after treatment with bevacizumab, and as

can be seen, this drug significantly decreased the individual's Cho/NAA ratio and significantly raised their Cr levels [10], illustrating the drug's efficacy as treatment and establishing its discovery.

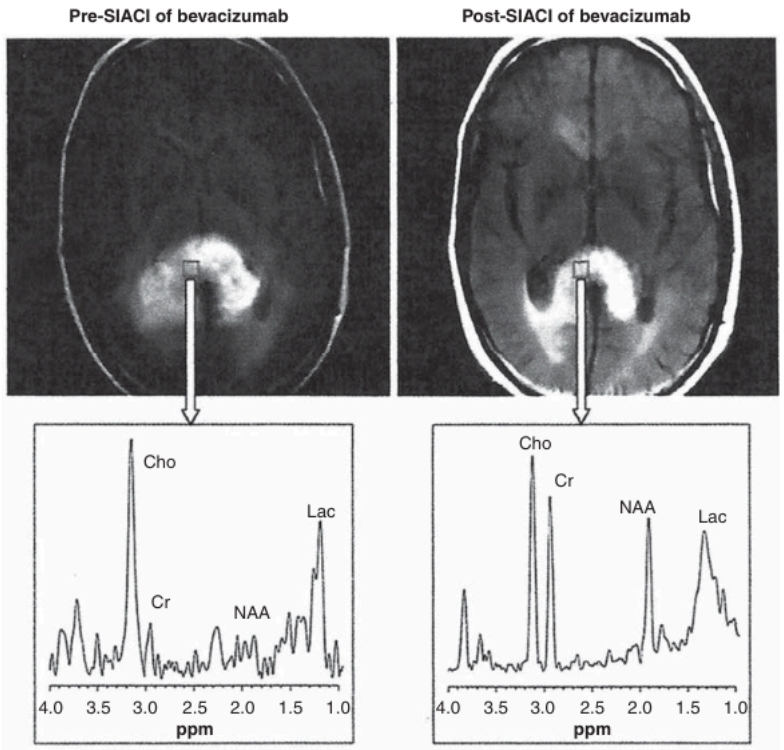

Fig. 15: 1H-MR spectra for a voxel in a GB the splenium of the corpus collosum before (A) and after (B) administration of bevacizumab, demonstrating reduction of the CNI (tCho/NAA ratio) post-treatment. [10, Fig. 1]

Food Science:

Another industry which utilizes the likes of MRS is that of food sciences. Specifically, due to the NMR abilities posed by MRS, it is able to be used for quality control of foods just like that of pharmaceuticals. By using MRS, researchers are capable of analyzing the nutrients within foods to authenticate their contents and maintain the integrity of food labels in industry. This is significant in maintaining the

safety of the general public and preventing contaminated foods from reaching the shelves. Due to the non-invasive nature of MRS, products do not have to be damaged in any way in order to go through MRS testing, making it an effective quality control method as well as an accurate one. As depicted in Figure 16, sequential imaging and MRS were utilized to examine the crystallizing oil-in-water emulsion. The emulsion is observable in both the imaging and the merging of peaks in the spectrum [7]. By performing studies like these, researchers can determine the behaviors and characteristics of the oil, and determine its safety and shelf life for consumption purposes. In this way, MRS can also be utilized to determine the shelf life of certain foods, especially perishable foods which have a distinct decay progression that is easily identified through MRS biomarkers.

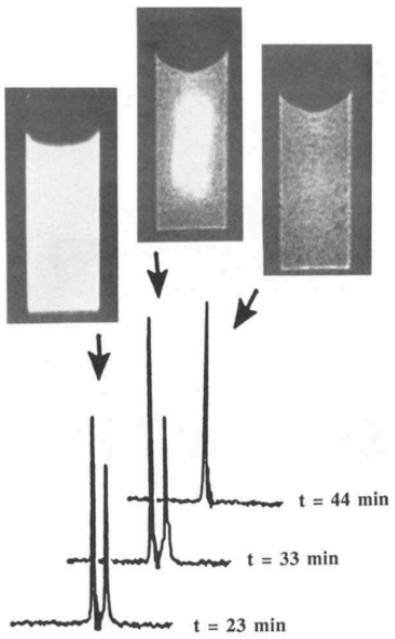

Fig. 16: Sequential imaging and localized spectroscopy of a crystallizing oil-in-water emulsion. [7, Fig. 6]

Forensics:

One last alternative industry which employs the wonders of MRS technology is that of forensics, mainly for identifying the age of postmortem tissues. Based on spectral data returned through MRS on deceased tissue, researchers are able to analyze the quantities of specific biomarkers in order to make an accurate prediction as to how long that tissue has been deceased for. Looking at Figure 17, magnetic resonance spectra of a sheep's brain can be seen. When the sheep was alive, there were many biomarkers that were not present in the spectrum that after 3 days postmortem, were extremely prevalent. Similarly, after 15 days postmortem, some biomarkers were so highly concentrated within the brain tissue that the spectrum had to be magnified by x0.05 in order to capture the data. Due to the immense differences in spectral data over such short periods of time, researchers are able to distinctly identify how long tissues have been deceased for and can accurately predict these time periods to the hours, making it a very useful tool for investigative purposes. Additionally, MRS can be employed on postmortem tissue to identify specific chemical signatures which may indicate cause of death, such as metabolic disorders or chemical signatures of specific toxins/poisons [8]. Although this might be the case, there are often alternative methods that are used for cause of death identification, making MRS not a primary avenue in this division of forensics.

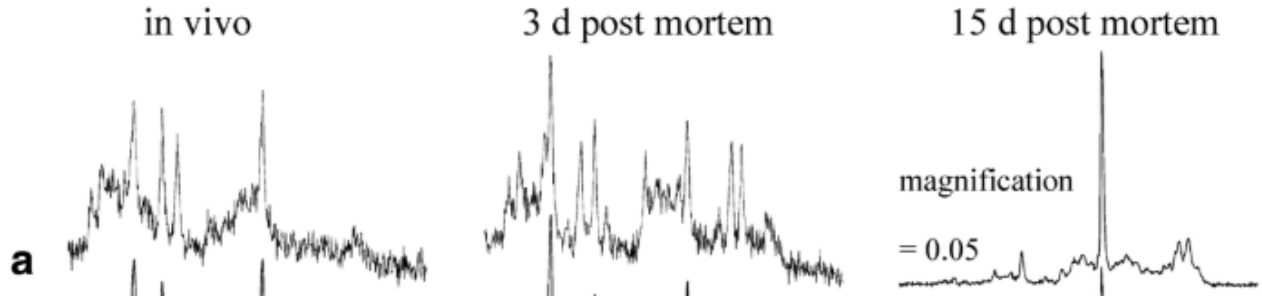

Fig. 17: a) Three spectra of sheep brain acquired in vivo, 3, and 15 days postmortem, respectively [8, Fig. 1]

Challenges of MRS:

Although MRS is such a successful technology, there are several challenges and limitations that come along with it just like any other great technology. MRS inherently has a poor signal to noise ratio (SNR), as the areas of tissue being analyzed are so localized, and the RF signal from the low-concentration metabolites are so small, the machines do not effectively pick up enough of the signal and consequently have much more noise per magnitude of signal. To counteract this challenge, engineers have deployed multi channel receiver coils on MRS machines, which essentially act as “extra ears” for the machine in picking up RF signals [2]. By having additional coils picking up more of the very small metabolite signals, a stronger and more accurate signal is able to be produced by the MRS, consequently raising the SNR.

Another challenge faced with MRS is B_0 inhomogeneity. Due to there being so many different tissue materials and the presence of elements such as air within a spectroscopy, the magnetic field easily becomes uneven in strength throughout the imaging volume. This leads to issues such as poor image quality and inaccurate spectrum data, which counteracts the benefits of MRS studies. In order to negate the issues from B_0 inhomogeneity, engineers use shimming coils in the x y and z directions in order to repair these imbalances and “smooth out” the bumps and valleys [2]. In doing so, shimming coils allow for repaired image resolution and accurate spectral data, both of which are critical to the benefits of MRS.

Motion correction is also an important solution MRS employs for imaging and spectroscopy on those who cannot stay still, such as infants, children, and animals. Due to their constant movement during the imaging and spectroscopy process, imaging and spectrum analysis can become distorted due to the positions of localization within the magnetic field becoming misaligned, and the voxels will shift relative to the regions of interest within the body. This leads to inaccurate imaging and metabolite concentration data, but motion correction can be applied using image based navigators [2]. These navigators are used to correct magnetic field localization and voxel shifts, repairing distorted images as if the area of interest was relatively still. In doing so, engineers are also able to correct the metabolite spectrum data that also became inaccurate due to motion.

Finally and probably one of the most critical challenges when performing MRS is that engineers need to efficiently balance echo time (TE), repetition time (TR), and field strength during operation. While T1 and T2 are inherent to the specific metabolites being worked with, TE and TR need to be properly adjusted based on the scope of the specific experiment being conducted by engineers and what their objectives are. Looking back at Figure 6 for example, let's say that that specific researcher's objective was to detect all of the biomarkers in the given list. Although the intermediate TE provides more accurate data and less baseline distortion, the intermediate TE is unable to pick up the signals from all of the objective metabolites. For this reason, the engineer must give and take with his spectral accuracy in order to obtain more data. By decreasing his TE to 35 ms and consequently decreasing his TR, although the baseline distortion is much greater, the engineer is able to collect significantly more signal

samples, and is able to successfully detect all of the desired metabolites [2]. This illustrates that there is not a single universal solution or approach when working with MRS, and engineers need to keep in mind the scope of their project in order to accurately determine what field strength and TE and TR they desire to operate with.

Future Directions of MRS:

In regards to the potential forward horizons for MRS, one of the main areas of study which is expected to gain significant traction in the next several years is that of liver disease. Liver disease and liver-related issues have become a prevalent issue in society due to modern lifestyles and diets, making it an avenue of medical concern where major research and money is being contributed. Liver disease has very few available metabolic profiling techniques currently in use or even in research development, making it prime real estate for future MRS discovery.

Another avenue of future development falls in the development of high field in-vitro MRS techniques to be employed in bio-fluid studies. In-vitro MRS is performed in laboratory settings, and currently there are many limitations and issues that arise with employing high magnetic fields in these environments outside of the body. Due to the difficulties in picking up the signals of very low concentrated metabolites in-vitro, high field in-vitro MRS is currently not a significant technology in industry practice as it is simply not feasible. Should these limitations be overcome though, high field in-vitro MRS does present very promising benefits in high sensitivity and detection ability, making it a technology of the future [4].

Speaking of technologies of the future, PET/MRS is also a significant technology which is very similar to that of PET/MRI and PET/CT machines which exist in all hospitals and clinics in the modern world. By combining PET and MRS, acquisition of specific sets of metabolic events and functional data would be able to be collected simultaneously, with improved spatial resolution and sensitivity to match [9]. The main goal of engineers in technology development is efficiency, and if PET/MRS will be able to provide avenues of MRS that are not only faster but also more accurate, PET/MRS may just become the gold standard for MRS procedures in the near future.

Conclusions:

All in all, magnetic resonance spectroscopy (MRS) is an absolutely indispensable technology which has shaped modern society and drastically improved overall quality of life. Its non-invasive nature, combined with its versatility and relatively fast acquisition times to alternative methods, positions it as a formidable tool in medical diagnostics. With its ability to swiftly provide invaluable metabolic information and insights into tissue composition, MRS offers high diagnostic value across various medical domains, including neurological disorders, metabolic diseases, psychiatric diseases, tumors/cancer, and many other sectors. Moreover, MRS extends its reach beyond clinical applications, finding relevance in research endeavors and alternative industries such as pharmaceuticals, food science, and forensics, giving MRS significant value beyond that of diagnostics. As a cornerstone of modern medical technology, MRS not only drives advancements in healthcare but also inspires the next generation of young

engineers, as it is surreal technologies such as these that captivate the minds of those who aspire to become engineers.

Works Cited

[1] Christoph Juchem, Douglas L. Rothman, “Chapter 1.1 - Basis of Magnetic Resonance”, *Magnetic Resonance Spectroscopy*, Academic Press, (2014); 3-14.

[2] Eva-Maria Ratai, R. Gilberto González, “Chapter 5 - Clinical Magnetic Resonance Spectroscopy of the Central Nervous System”, *Handbook of Clinical Neurology*, Elsevier, Volume 135, (2016); 93-116.

[3] T. Mikkelsen, D. Hearshen, “CHAPTER 28 - Magnetic Resonance Spectroscopy”, *Handbook of Neuro-Oncology NeuroImaging*, Academic Press, (2008); 257-263.

[4] Tognarelli, Joshua M., et al. "Magnetic Resonance Spectroscopy: Principles and Techniques: Lessons for Clinicians", *Journal of Clinical and Experimental Hepatology 5.4* (2015); 320-328.

[5] Zhu H, Barker PB. “MR Spectroscopy and Spectroscopic Imaging of the Brain”, *Methods Mol Biol*. (2011); 711: 203-26.

[6] Kohut, Stephen J., and Marc J. Kaufman. "Magnetic resonance spectroscopy studies of substance use disorders: Current landscape and potential future directions." *Pharmacology Biochemistry and Behavior* 200 (2021): 173090.

[7] Simoneau, C., M. J. McCarthy, and J. B. German. "Magnetic resonance imaging and spectroscopy for food systems." *Food Research International* 26.5 (1993): 387-398.

[8] Ith, Michael, et al. "Observation and identification of metabolites emerging during postmortem decomposition of brain tissue by means of in situ 1H-magnetic resonance spectroscopy." *Magnetic Resonance in Medicine: An Official Journal*

of the International Society for Magnetic Resonance in Medicine 48.5 (2002): 915-920.

[9] Lin, Gigin, and Yuen-Li Chung. "Current opportunities and challenges of magnetic resonance spectroscopy, positron emission tomography, and mass spectrometry imaging for mapping cancer metabolism in vivo." *BioMed Research International* 2014 (2014).

[10] Ross, Brian D. "High-field MRS in clinical drug development." *Expert Opinion on Drug Discovery* 8.7 (2013): 849-863.